Sensorless Control of PMSM Based on Low Frequency Voltage Injection at Low Speeds and Standstill

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***Abstract*—Most sensorless control methods have problems when they are used in surface-mounted permanent magnet synchronous motor (PMSM) that has little salient effect. To solve this problem, a novel low frequency voltage injection method is developed in this paper. Low frequency rotating voltage vectors are imposed on the stator, and the current responses are analyzed to get the rotor position information. Different from high frequency voltage injection, electric torque produced by low frequency voltage would has enough time to move the rotor a little but not big enough to rotate the rotor. The electric torque, friction torque and the cogging torque together will make the rotor oscillate around an equilibrium point. The back electromotive force produced by the rotor movement would have effect in the current response, which is finally used in the rotor position estimation. A rotor position estimation structure based on phase lock loop (PLL) is designed to continuously estimate the rotor position at low speeds and some offline experiments and compensation are added to get more accuracy estimation results. Experiments have been finished on a 1.25 kW PMSM at low speeds and standstill, the correctness of the proposed method has been demonstrated.**

Keywords—sensorless control; low frequency voltage injection; permanent magnet synchronous motor (PMSM); phase lock loop (PLL)

I. INTRODUCTION

During recent years, much attention has been paid on sensorless control of permanent magnet synchronous motor (PMSM). In some literature, sensorless control has been realized with direct torque control (DTC) [1]. In addition to DTC, there are mainly two kinds of sensorless control methods.

One sort of well-developed methods is based on back electromotive force (EMF), which can be estimated by using state observer [2], extended Kalman Filter (EKF) [3] or sliding mode controller [4], etc. Even if the successful use of these methods in medium and high speeds, they meet barriers at low speeds and standstill, as the back EMF would be quite low or even zero [5]. Meanwhile, another mature kind of methods using high frequency signal injection has been developed, which is based on saliency effect and can be used at low speeds [6][7]. In order to get the information of the rotor position more precisely, some literature has even considered motor design together with sensorless control [8].

On the other hand, for surface mounted motor with little saliency, those high frequency injection methods may fail. To solve this problem, a low-frequency signal injection method is used [9-11], in which pulsating voltage vectors are used.

In this paper, a novel low frequency signal injection method using rotating voltage vectors is proposed, which can be used at low speeds and standstill with any rotor type.

II. LOW FREQUENCY VOLTAGE INJECTION UNDER STANDSTILL CONDITION

A. Mathematic Model of PMSM

The common used mathematic model of PMSM can be written as below.

$$u_d = Ri_d + L_d di_d / dt - \omega_r L_q i_q \tag{1}$$

$$u_q = Ri_q + L_q di_q / dt + \omega_r (L_d i_q + \psi_{PM}) \tag{2}$$

$$T_e = 2p[(L_d - L_q) i_d i_q + \psi_{PM} i_q] / 3 \tag{3}$$

$$d\omega_r / dt = p(T_e - T_l - T_f - T_c) / J \tag{4}$$

$$d\theta_r / dt = \omega_r \tag{5}$$

Where u_d, u_q are the dq-axes voltage, i_d, i_q are the dq-axes current, R is the motor resistance, L_d, L_q are the dq-axes inductance, ω_r is the rotor speed, T_m, T_l, T_f and T_c are respectively the electric torque, load torque, friction torque and cogging torque, p is the pole pairs, J is the rotational inertia, ψ_{PM} is the permanent magnet flux linkage, and θ_r is the rotor position.

B. Current Response of PMSM inder Low Frequency Voltage Injection

The voltage is directly imposed on the motor through the inverter using the signal form in (6) that is similar with the one in high frequency signal injection method, but the frequency is

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low. If the injection frequency is high, the produced torque is also high frequency and the rotor may stand still with zero back EMF. The current response of the motor can be expressed in (7) with dq-axes inductance equal each other ($L=L_d=L_q$).

$$u_{\alpha\beta} = U_{amp} e^{j\omega t} \tag{6}$$

$$i_{\alpha\beta} = I_{amp} e^{j(\omega t - \varphi)} \tag{7}$$

Where,

$$I_{amp} = U_{amp} / \sqrt{R^2 + \omega^2 L^2}$$

$$\varphi = \arctan(\omega L / R)$$

Where αβ-axes are the two-phase static coordinate system, $u_{\alpha\beta}$ is αβ-axes voltage, U_{amp} is the amplitude of the rotating voltage vector, and ω is the angular frequency of the injection signal.

Under low frequency and proper voltage amplitude, the produced electric torque would have enough time to move the rotor a little in every cycle. However, the electric torque is not big enough to overcome the resistance torque (friction torque and cogging torque) to rotate the rotor, and the rotor would oscillate around an equilibrium point. The phase relations of some important motor quantities are shown in Fig. 1 and only the fundamental harmonic is considered. The electric torque should have the same phase with the q-axis current, the friction torque is in phase with the rotor speed, and the cogging torque is in phase with the rotor position. During the stage I in Fig. 1, the electric torque (T_e) is smaller than the resistance torque (T_f+T_c), so the rotor speed would decelerate. While in stage II, the rotor speed would accelerate.

With proper frequency and amplitude, the q-axis current would be about 90° ahead the rotor speed. The back EMF caused by the oscillation would affect the current response. The voltage and current vector diagram can be shown in Fig. 2(b) compared with the one in Fig. 2(a) when the rotor keeps still. In Fig. 2(b) the injection frequency is low, so the voltage drop on the inductance is neglected. According to the diagram in Fig. 2(b), the current response of αβ-axes would be ellipse as is shown in Fig. 3 with expression in (8) and the long axis of the ellipse would be the rotor position. Fourier analysis can be used to calculate the rotor position in (8) precisely at the injection frequency.

$$i_{\alpha\beta} = i_{dq} e^{j\theta_r} = I_A e^{j\omega t} + I_B e^{-j(\omega t - 2\theta_r)} \tag{8}$$

The polarity determination can be gotten from the current response, as is shown in Fig. 4. The amplitudes of current response at point C and point D are different due to the asymmetrical oscillation process in one cycle. The current vector rotates clockwise, while the rotor position oscillates back and forth. Under this condition, the torque produced by the current vector would be asymmetrical, and the acceleration and deceleration process will be different.

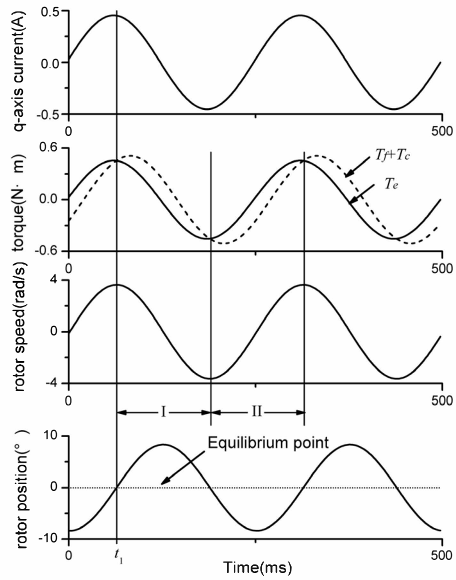
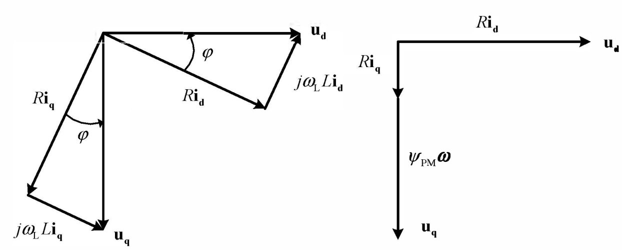


Fig. 1. Phase relationship between motor quantities under low frequency voltage injection.



(a) Rotor at standstill condition (b) Rotor at oscillation condition

Fig. 2. Voltage and current vector diagram under signal injection.

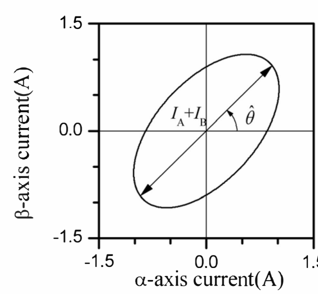


Fig. 3. Current responses under low frequency injection.

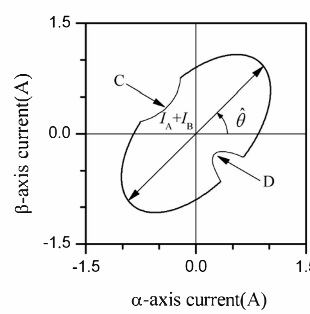


Fig. 4. Current responses under low frequency injection for polarity determination.

III. Low Frequency Voltage Injection under Low Speeds

Under standstill condition, the injected voltage in (6) is realized by the inverter directly. While under low speeds condition, additional current control loop is needed to control the dq-axes currents. The Proportional Integral (PI) controller in the control loop would affect the current response.

On the other hand, as the rotor rotates, the estimation rotor position should be fast enough to track the changing rotor position.

A. The effects of PI controller

The block diagram of current control loop with low frequency voltage injection is shown in Fig. 5, where u_{dL} indicates the injection voltage.

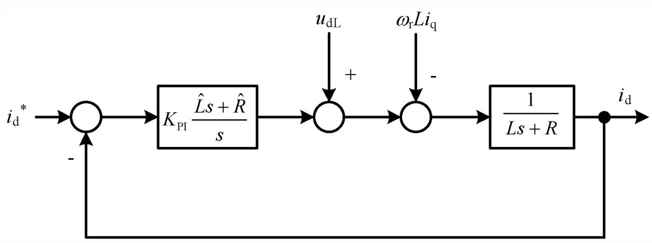


Fig. 5. Low frequency voltage injection with PI controller.

Suppose that the parameters used in the PI controller are accuracy, the transform function from u_{dL} to i_d can be written as below.

$$\frac{I_d(s)}{U_{dL}(s)}=\frac{s}{(Ls+R)(s+K_{PI})} \tag{9}$$

When the injection voltage is directly imposed on the motor without PI controller, the transform function would be (10).

$$\frac{I'_d(s)}{U_{dL}(s)}=\frac{1}{Ls+R} \tag{10}$$

Compare (9) and (10), the effects of PI controller on current response under low frequency voltage injection can be written in (11).

$$\begin{aligned}&\frac{I_d(s)}{I'_d(s)}=I_{PI}\mathrm{e}^{-\mathrm{j}\theta_{PI}}\\ &\theta_{PI}=\pi/2-\arctan(2\pi f/K_{PI})\\ &I_{PI}=2\pi f/\sqrt{(2\pi f)^2+K_{PI}^2}\end{aligned} \tag{11}$$

The phase difference (θ_{PI}) would bring error to the final estimation result, which should be compensated, while the reduced amplitude would make the information hard to observe. To guarantee enough amplitude of the current response, the amplitude of the injection voltage should be increased, or the K_{PI} in PI controller should be decreased. In fact, the latter is used to make dynamic process slow, because fast dynamic process would make the low frequency injection signal hard to extract.

B. Rotor position estimator under low frequency voltage injection

Considering the effects of PI controller, the current response under low frequency voltage injection with PI controller can be written as following.

$$i_{\alpha\beta}=I_{PI}(I_A\mathrm{e}^{\mathrm{j}(\omega t-\theta_{\omega1}-\theta_{PI})}+I_B\mathrm{e}^{-\mathrm{j}(\omega t-2\theta_r-\theta_{\omega2}-\theta_{PI})}) \tag{12}$$

As the estimation rate is slow using Fourier analysis, a rotor position estimator is designed based on PLL in Fig. 6.

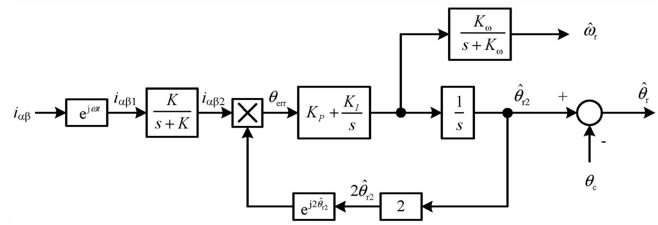


Fig. 6. Rotor position estimator based on PLL.

The transform is done first in Fig. 4 using (13).

$$\begin{aligned}i_{\alpha\beta1}&=i_{\alpha\beta}\mathrm{e}^{\mathrm{j}\omega t}\\ &=I_{PI}(I_{\mathrm{A}}\mathrm{e}^{\mathrm{j}(2\omega t-\theta_{\omega1}-\theta_{PI})}+I_{\mathrm{B}}\mathrm{e}^{\mathrm{j}(2\theta_r+\theta_{\omega2}+\theta_{PI})})\end{aligned} \tag{13}$$

As the frequency of the current component containing rotor position information is quite slow in (13), a low-passed filter is used to get it. At steady condition, the $i_{\alpha\beta2}$ would have the form in (14).

$$\begin{aligned}&i_{\alpha\beta2}=I_{B2}\mathrm{e}^{\mathrm{j}(2\theta_r+\theta_{\omega2}+\theta_{PI}-\theta_{\omega3})}\\ &I_{B2}=I_{PI}I_BK/\sqrt{\omega_r^2+K^2}\\ &\theta_{\omega3}=\arctan(\omega_r/K)\end{aligned} \tag{14}$$

And (14) can be expressed as (15).

$$\begin{aligned}&i_{\alpha\beta2}=I_{B2}\mathrm{e}^{\mathrm{j}2\theta_{r2}}\\ &\theta_{r2}=\theta_r+(\theta_{\omega2}+\theta_{PI}-\theta_{\omega3})/2\end{aligned} \tag{15}$$

The rotor position error is obtained by cross product operation in (16).

$$\theta_{err}=i_{\alpha\beta2}\times\mathrm{e}^{\mathrm{j}2\hat{\theta}_{r2}}=I_{PI}I_B\sin 2(\theta_{r2}-\hat{\theta}_{r2}) \tag{16}$$

Using PLL structure, the estimation value of θ_{r2} can be obtained, and the compensation component θ_c should have the form as below.

$$\theta_c=(\theta_{\omega2}+\theta_{PI}-\theta_{\omega3})/2 \tag{17}$$

θ_{PI} can be calculated by (11), $\theta_{\omega2}$ is the errors under standstill condition without PI controller, and $\theta_{\omega3}$ should be calculated by (14).

C. Parameters settings for low frequency voltage injection

Low frequency voltage injection method is designed to be used under standstill and at low speeds, and the speed range is set as 0~6.28rad/s, with electric frequency 0-1Hz.

In order to get rid of the effects of the fundamental harmonic on the injection voltage, the injection frequency should be much larger than the rotor speed, while the injection frequency should be low enough to make the rotor swing. In this paper the injection frequency is f=50Hz, with amplitude U_s=4V, while the parameter in PI controller is K_{PI}=10.

For the $i_{\alpha\beta 1}$ in Fig. 6, the fundamental harmonic is not contained in (13), as the frequency of it would be about 50Hz after using (13), which should be filtered by the low-passed filter. And the parameter in the low-passed filter is set as K=10.

As the rotor speed is low, the parameters of the PI controller in the PLL can be chosen in a wide range, and they are set as K_P=1000, K_I=100. The speed filter parameters K_ω=10.

Based on the parameters, the compensation errors in (17) are calculated respectively in (18) and (19).

$$\theta_{PI} = \pi/2 - \arctan(2\pi f / K_{PI}) = 0.31 \tag{18}$$

$$\theta_{\omega 3} = \arctan(\omega_r / K) = 0.1\omega_r \tag{19}$$

The $\theta_{\omega 2}$ is determined by off-line experiments without PI controller. Using injection frequency f=50, and amplitude U_s=4V, the estimation errors are shown in Fig. 7 and the average error is used, as is shown in (20).

$$\theta_{\omega 2} = 0.57 \tag{20}$$

Finally, the compensation component can be written as below.

$$\theta_c = 0.44 - 0.05\omega_r \tag{21}$$

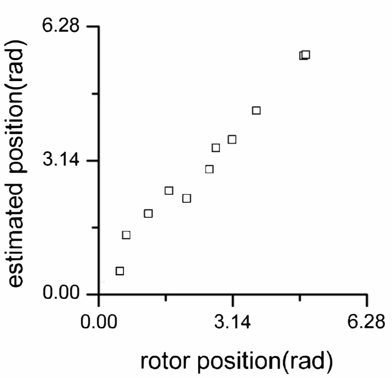


Fig. 7. Estimation results under standstill condition (f=50Hz, U_s=4V).

D. Hybrid sensorless control stucture based on low frequency voltage injeciton

Due to the constraint of the low frequency, the low frequency voltage injection method can only be used during start-up and low speeds. When the speed rises to a certain value, other sensorless methods should be applied. In Fig. 8, a hybrid sensorless control structure is designed by combining the proposed low-frequency voltage injection method with sensorless method based on luenberger observer [12].

When the speed is low than the set value (6.28rad/s), the outputs of low-frequency voltage injection method are used, while at speed higher than the set value, the estimation results of the luenberger observer are used. At low speeds, the luenberger observer is running and ready for the switchover, while at high speeds the low frequency voltage injection method is stopped.

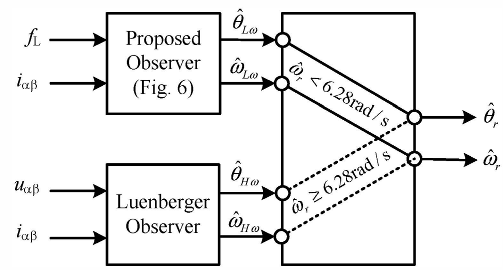


Fig. 8. Hybrid position sensorless control structure.

The hybrid sensorless control method can be used in PMSM control system and the block diagram is shown in Fig. 9. The frequency and amplitude of the injection voltage are set as f=50Hz, U_s=4V. The parameters in the PI controller are determined by the estimated rotor speed. At low speeds (<6.28rad/s), the low frequency voltage is injection with K_{PI}=10, while at higher speeds, the injection is stopped with normal PI parameters (K_{PI}=1000).

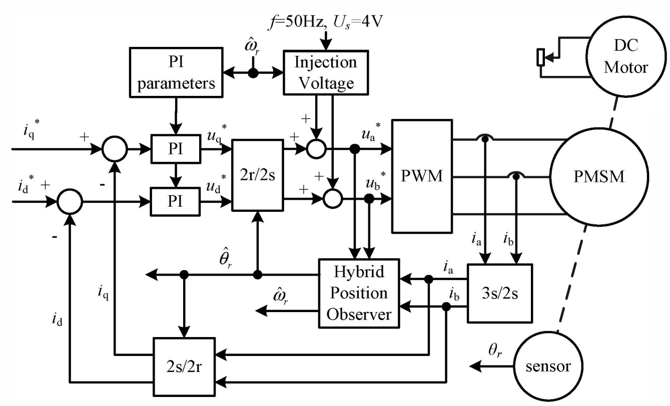


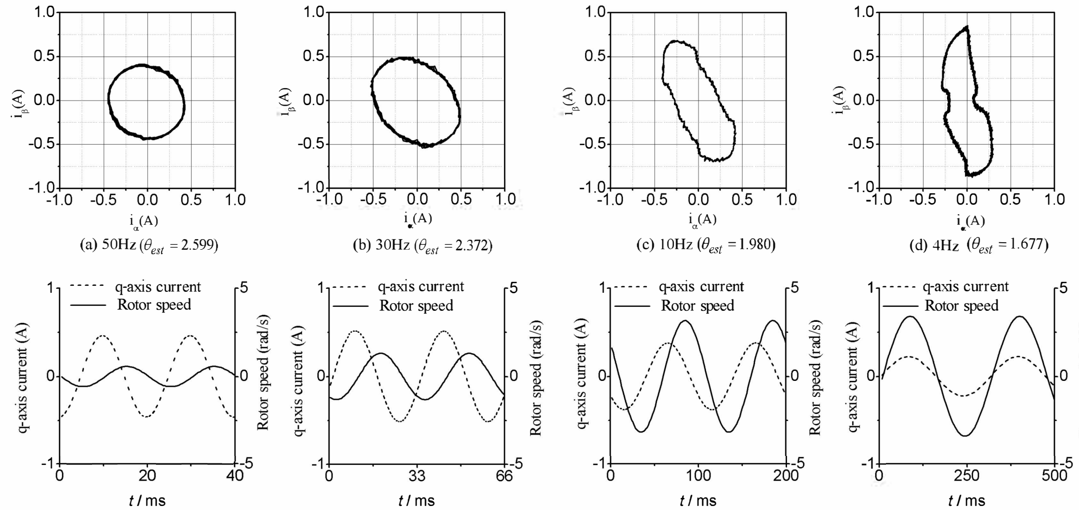
Fig. 9. Hybrid position sensorless PMSM control system.

IV. Experimental Results

Experiments are finished on a PMSM platform, with parameters in table. 1.

TABLE 1 PMSM PARAMETERS

Parameters	Values
P_n	1.25 kW
V_n	195 V
p	2
R	2.16 Ω
L_d	8 mH
L_q	8 mH
ψ_{PM}	0.1487 Wb



(a) 50Hz (θ_{est} = 2.599) (b) 30Hz (θ_{est} = 2.372) (c) 10Hz (θ_{est} = 1.980) (d) 4Hz (θ_{est} = 1.677)

(e)50Hz (100°) (f)30Hz (103°) (g)10Hz (70.3°) (h)4Hz (5.65°)

Fig. 10. Experimental results of low frequency voltage injection method with different injection frequencies at standstill condition (θ_r=1.65).

A. Experimental results under standstill condition

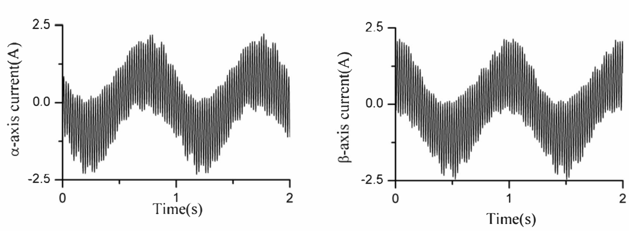
Rotating voltage vector is directly imposed on the motor through the inverter with different frequencies and proper amplitude (U_s=2V). The current responses are shown in Fig. 10(a)(b)(c)(d), while the estimated rotor position is calculated using Fourier analysis. The phase relationship between q-axis current and rotor speed is shown in Fig. 10(e)(f)(g)(h), in which only the fundamental harmonic is considered.

The estimated rotor position is more precise at low frequency in Fig. 10(d), when the q-axis current is nearly the same phase with rotor speed at this injection frequency (Fig. 10(h)). Estimation errors are shown at higher injection frequency, which is due to the phase difference between q-axis current and the rotor speed.

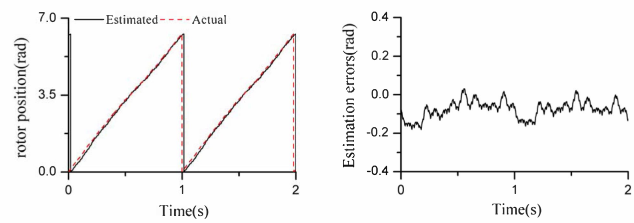
B. Experimental results of low frequency voltage injetion

Experiments are done based on Fig. 6, the steady and dynamic performances are tested.

Steady experimental results are shown in Fig. 11 with current command i_q^*=1A, i_d^*=0. The rotor speed is about 6.28rad/s by adjusting the load. As is shown in the figure, the oscillation in αβ-axes current is about 1A due to the impact of the injection voltage, and the rotor position estimation errors are small, the control system can run reliably.



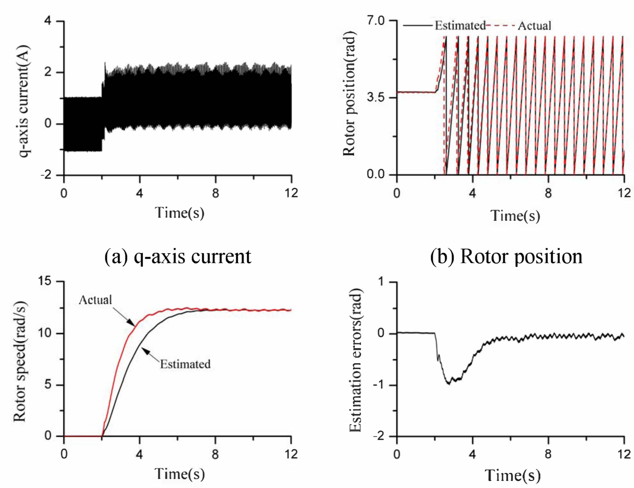
(a) α-axis current (b) β-axis current



(c) Rotor position (d) Estimation errors

Fig. 11. Experimental results of the low frequency voltage injection method at steady state.

Dynamic experimental results are shown in Fig. 12 with step command on q-axis current. As is shown in the figure, the estimation error is large at start-up, which is determined by the dynamic performance of the proposed method. In addition, the filtered estimation speed lags the actual one.



(a) q-axis current (b) Rotor position

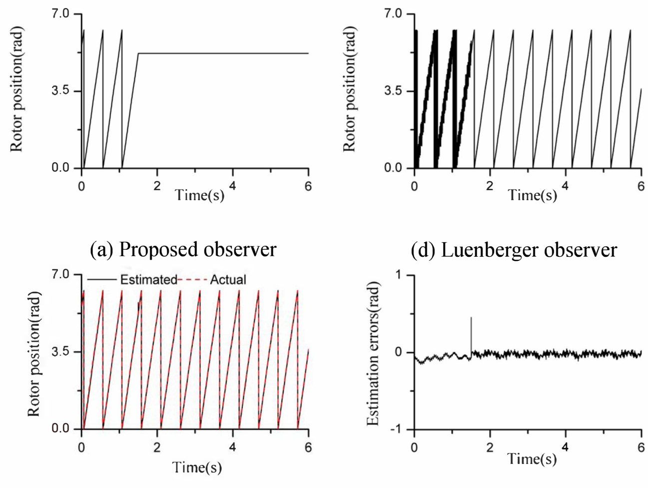
(c) Rotor speed (d) Estimation errors

Fig. 12. Experimental results of the low frequency voltage injection method during start-up.

C. Experimental results of hybrid sensorless control method

The hybrid sensorless control method in Fig. 8 is tested using the control system in Fig. 9.

First, the switchover experiments are done with speed at 12.52rad/s, the results are shown in Fig. 13. As is shown in the figure, the estimation results of luenberger observer contain harmonics under low frequency voltage injection, while after the switchover, the harmonics in the estimation results vanished as the injection is stopped. At the switch point the estimation errors are quite big due to the harmonics in the luenberger observer, however, this shows little effect on the control system.

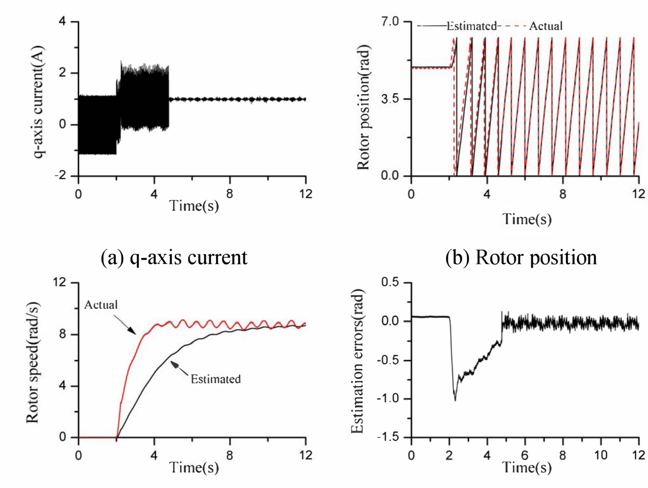


(a) Proposed observer (d) Luenberger observer

(c) Hybrid observer (d) Estimation errors

Fig. 13. Steady experimental results of hybrid sensorless control structure.

Dynamic performance of the hybrid sensorless control method is tested. A step command of 0-1A is given on q-axis current, and the results are shown in fig. 14. The switch process is smooth shown in the figure, and after the switch point, the injection is stopped and the harmonics in the q-axis current is sharply decreased.



(a) q-axis current (b) Rotor position

(c) Rotor speed (d) Estimation errors

Fig. 14. Dynamic experimental results of hybrid sensorless control structure.

Furthermore, the estimation error peak at the switch moment, which is the same situation in Fig. 13. As the filter with low cut-off frequency is used in the speed estimation, the switch process is smooth with switch condition using the estimation speed. Although the estimation speed lags the actual one, the low frequency voltage injection method can be run at higher speed (<15rad/s), the proposed method can be run reliably under PMSM control system with poor dynamic performance.

V. CONCLUSIONS

A low frequency voltage injection method is proposed in this paper for PMSM sensorless control. The rotor micro-movement is used to generate back-emf, which would affect the current response that is finally used in the rotor position estimation. Fourier analysis is used at standstill, and a position estimator is designed based on PLL for low speeds application. The low frequency voltage injection method can combine with other sensorless methods to estimate the position at whole speed range.

The proposed method can run reliably at low speeds with high estimation accuracy. Nevertheless, the torque is small and the dynamic performance is sacrificed.

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